neutron are utilized, whereas in the helium nucleus all valencies are saturated.

It is therefore to be expected in general that the preferred nature of even numbers will manifest itself in a particularly high stability of the nuclei which contain both an even number of protons and an even number of neutrons, which we shall call ‘doubly even nuclei’. Those nuclei in which either N or Z is an even number while the other one is odd, are less stable; those in which both N and Z are odd, are still less stable. There is a general empirical proof of this statement: It might be expected that the more stable a certain nucleus is, the more frequently it would occur in nature, since at the time of the original formation of nuclei from their building blocks, the most stable nuclei would have a preferred status both as to frequency of formation and the chance of remaining intact. Many years ago, Harkins attempted to determine the empirical relationships between odd or even numbers on the one hand and the natural abundance of elements on the other. He found that the by far most abundant atomic species were, in fact, those which today we know to have ‘doubly even’ nuclei. Nuclei in which either N or Z is an odd number—‘odd nuclei’—are much rarer, and rarest of all are the nuclei in which both N and Z are odd numbers, which we shall call ‘doubly odd’ nuclei.

‘Doubly even’ oxygen, ${}_8O^{16}$, is one of the most common elements. The ‘odd’ lithium, ${}_3Li^7$, is much rarer. Finally, we must note the fact that only a very few stable ‘doubly odd’ nuclei exist at all. The simplest one of these is the deuteron, the nucleus of deuterium, ${}_1D^2$. The only others are: the lithium nucleus ${}_3Li^6$, the boron nucleus ${}_5B^{10}$, and the nitrogen nucleus ${}_7N^{14}$. All the others of this kind are radioactive and change (‘decay’) by the emission of electrons or positrons.

On the ground of the foregoing considerations, let us now analyse the question of nuclear stability in greater detail. We have already described the energy surface (Figure 12). This is a very steeply slanting surface with a channel or groove, at the bottom of which the stable nuclei lie. We shall now make a cut through the surface, from the upper left to the lower right, at an angle of 45 deg. to the axes, diagonally to the channel, so giving us a cross-section of the surface (Figure 18). As a result of the nature of this cut, this cross-section will contain only nuclei with the same $N + Z$ total, i.e. with the same mass number. We arrange this cross-section first of all so that $N + Z$ is an odd number. The result is a curve, the lowest point of which is on the valley floor. Only those nuclei lie on this curve which are capable of changing into each other by the emission of an electron or positron; only the lowest nucleus, the nucleus possessing the greatest binding energy, is expected to be stable. In this diagram, the neutron numbers (their absolute magnitudes are of no importance here) are recorded on the abscissa in ascending order, while the binding energy is shown by the ordinate. Since $N + Z$ is constant, an increase